llmXive Automated Review of Data Journalist Agent: Transforming Data into Verifiable Multimodal Stories

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific factual claims regarding model versions, future events, and statistical counts that are not supported by the provided evidence or are factually impossible.

First, in **appendix/0_setup.tex** (Table 1), the authors list specific API models such as `openai/gpt-5.4-image-2` and `gpt-5.5-xhigh`. These model versions do not exist in the public domain as of the current date (the paper is dated June 2026, but the models listed appear to be hallucinated future versions or typos for existing models like GPT-4o). Citing non-existent models as the backbone of the experimental results undermines the verifiability of the entire study. The authors must correct these to the actual models used (e.g., GPT-4o, Claude 3.5/4) or provide a valid link to a private/beta API if these are real but unreleased models.

Second, in **sec/3_discovery.tex**, the paper describes the "ArXiv submissions" case study with the claim: "arXiv stopped treating an institutional email as enough to endorse a first-time submitter in January 2026." Since the paper is being reviewed in 2026 (implied by the date on the arXiv ID and the content), presenting a specific policy change in "January 2026" as a completed historical fact is highly suspicious. If the paper is written *in* 2026, this is a future event relative to the current real-world time, or a hallucination of a specific date. If this is a projection, it must be explicitly framed as a prediction or a hypothetical scenario, not a reported fact. If it is a real event, a citation to the specific arXiv policy announcement is required.

Third, in **sec/5_experiments.tex** (subsection "Computer-use agent as a cost-efficient alternative"), the text states: "almost every point (29/34) sits above the y=x line." However, the human study section explicitly states there were **53** reviewers. The denominator "34" is unexplained. It is unclear if this refers to a subset of articles, a specific condition, or if it is a calculation error. This discrepancy makes the statistical claim ambiguous and potentially inaccurate.

Finally, the bibliography contains a mismatch for `c-001` (NASA meteorite landings) which is flagged in the proofreader notes, though this is a minor citation error compared to the model hallucinations.

These issues require correction to ensure the claims accurately reflect the experimental reality and do not rely on hallucinated model versions or impossible timelines.

Action items.

- [**science**] In sec/3_discovery.tex, the claim that 'arXiv stopped treating an institutional email as enough to endorse a first-time submitter in January 2026' is temporally impossible as the paper is dated 2026 and this is a future event presented as a past fact. Verify if this is a hallucination or a projection that needs rephrasing as a prediction.
- [**science**] In appendix/0_setup.tex, Table 1 lists 'gpt-5.4-image-2' and 'gpt-5.5-xhigh' as API models. As of the current date, these model versions do not exist in public APIs (OpenAI's latest is the 4o series). These citations are factually incorrect and misrepresent the experimental setup.
- [**writing**] In sec/5_experiments.tex, the text states '29/34' points sit above the y=x line in Figure 5c, but the total number of data points in the scatter plot (n=53 reviewers) does not match the denominator 34. Clarify the sample size used for this specific correlation analysis.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively illustrates the system's workflow from raw data to a multimodal story with clear visual components. However, the caption contains grammatical errors and a raw URL embedded in the text that should be formatted more professionally.

Figure 2

The figure is a promotional image rather than a data visualization, failing to present the scientific evidence implied by the caption and the paper's context.

Figure 3

The figure provides a clear visual workflow of the Virtual Newsroom, but the caption is severely flawed with a truncated sentence and mathematical notation that contradicts the labels shown in the diagram.

Figure 4

Figure 4 effectively illustrates the 'Inspector' concept by visually mapping specific claims in a generated article to their underlying evidence (code snippets and reference URLs). The diagram is clear, the connections are logical, and it aligns perfectly with the caption's description of binding findings to code and reference evidence.

Figure 5

The figure effectively illustrates the three evaluation workflows with clear icons and flow, but the title and caption contain a grammatical error omitting the system name, and Panel A fails to visualize the overlap metric it claims to measure.

Figure 6

The figure is a bar chart of role contributions that completely mismatches its caption, which describes sentence and word statistics. The visual content and text are unrelated.

Figure 7

The figure presents a stacked bar chart of media counts across four categories, but the caption is critically incomplete ('Articles made by .'), failing to explain the x-axis labels or the specific comparison being made.

Figure 8

The figure presents a bar chart comparing two groups across six categories, but the caption is too vague to identify the categories, and the specific numerical values inside the bars lack clear definition.

Figure 9

The figure is visually clear with readable data and error bars, but the caption is generic and does not accurately describe the specific datasets shown on the x-axis.

Figure 10

The figure presents a bar chart with inconsistent units between the y-axis (percentages) and the data labels (decimals), creating ambiguity about the actual values. Additionally, the caption is too brief to fully explain the x-axis categories or the specific metric.

Figure 11

The bar chart is visually clear with data labels, but the y-axis metric and the specific definition of the roles on the x-axis are ambiguous without further context in the caption.

Action items.

- [**writing**] Figure 1: The caption contains a raw URL (https://osf.io/534g2/overview) embedded directly in the sentence structure, which is unprofessional and disrupts readability; this should be formatted as a citation or footnote.
- [**writing**] Figure 1: The caption text 'turns a raw dataset' lacks a subject (e.g., 'The agent turns...'), making the sentence grammatically incomplete.
- [**fatal**] Figure 2: The image is a promotional poster for the FIFA World Cup 2026, not a scientific figure. It lacks axes, data points, or statistical visualizations required to support the caption 'Sixteen Climates' or the paper's claims about data analysis.
- [**science**] Figure 2: The caption 'Sixteen Climates' is not supported by the visual content, which shows a single stadium scene. There is no data presented to substantiate the claim of '16 climates' or the temperature range mentioned in the image text.
- [**fatal**] Figure 3: The caption is truncated mid-sentence ('...those that ground the') and contains a dangling file reference '[pipeline.pdf]', indicating incomplete text.
- [**science**] Figure 3: The caption defines the Inspector's evidence set as $E = DRCFV$, but the diagram explicitly labels the Inspector's formula as $\mathcal{E} = \mathcal{D} \cup \mathcal{R} \cup \mathcal{C} \cup \mathcal{F} \cup \mathcal{V}$; the caption omits

the union operators present in the figure.

- [**science**] Figure 3: The caption states the Analyst emits results R with code-line provenance $RCDD$, but the diagram labels the Analyst's output simply as $\mathcal{R}, \mathcal{C}$, creating a contradiction in the variable definitions.
- [**writing**] Figure 5: The figure title and caption text contain a grammatical gap ('protocols for .') where the system name is missing.
- [**science**] Figure 5: Panel A depicts a 'Human-Agent Coverage' comparison but lacks a visual representation of the 'Opinion Overlap?' metric mentioned in the diagram, making the measurement process unclear.
- [**fatal**] Figure 6: The rendered image is a bar chart titled 'Individual role contribution' showing role percentages, but the caption describes 'Num. of sentences per article and Avg. words per sentence'. The visual content and caption are completely mismatched.
- [**science**] Figure 6: The figure displays data for 'Editor', 'Detective', 'Analyst', and 'Designer' roles, which contradicts the caption's claim of showing sentence counts and word averages per article.
- [**fatal**] Figure 7: The caption 'Articles made by .' is incomplete and grammatically broken, failing to identify the subject (likely the agent) or the metric being visualized.
- [**science**] Figure 7: The stacked bars show 'mean count per article' for specific media types (heading, interactive, audio, etc.), but the caption does not define these categories or explain what 'articles' are being compared (e.g., agent vs. human, or specific datasets).
- [**writing**] Figure 7: The x-axis labels ('Economist', 'Pudding', 'TidyTuesday') are ambiguous without context in the caption; it is unclear if these represent source publications, dataset types, or specific article examples.
- [**fatal**] Figure 8: The caption 'By rubric dimension' is insufficient; the x-axis labels ('1-Vis.', '2-Narr.', etc.) are undefined and do not match the rubric dimensions mentioned in Figure 5's caption.
- [**science**] Figure 8: The y-axis is labeled 'mean score (1-7)', but the bars contain small numerical labels (e.g., 4.17, 3.66) that are not explained; it is unclear if these are the exact means or raw data points.
- [**writing**] Figure 8: The legend uses color swatches ('Ours', 'Human') but does not explicitly state that blue corresponds to the agent and orange to the human, relying on visual inference.
- [**science**] Figure 9: The caption 'By source category' does not match the x-axis labels ('Economist', 'Pudding', 'TidyTuesday', 'Overall'), which represent specific datasets or an aggregate, not source categories.
- [**writing**] Figure 9: The legend is placed inside the plot area, reducing the visible space for the bars and error bars.

- [**science**] Figure 10: The y-axis is labeled 'Auditability' with a percentage scale (0-100%), but the bars contain decimal values (0.18, 0.28, 0.30, 0.25) that do not match the axis units (e.g., 0.18 vs 18%); the axis or the internal labels must be corrected to be consistent.
- [**writing**] Figure 10: The caption 'Human, per source' is insufficient; it should explicitly define what the x-axis categories (Economist, Pudding, TidyTuesday) represent and clarify the metric being measured.
- [**writing**] Figure 11: The y-axis label ('% of traced sentences citing role') is ambiguous; it is unclear if this represents the percentage of sentences where the role was cited, or the percentage of the role's output that was cited. The caption 'Individual role contribution' does not clarify this metric.
- [**writing**] Figure 11: The x-axis labels ('Editor', 'Detective', 'Analyst', 'Designer') are generic role names. The caption does not specify if these refer to the specific agents in the 'Virtual Newsroom' (Figure 3) or human roles, making the context of the contribution unclear.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology and coined phrases that may alienate non-specialist readers, particularly those outside the immediate field of agentic AI systems.

First, the term **"computer-use agent"** is introduced in the Abstract and Section 1 without definition. While the authors later describe it as an agent that "perceives the rendered interface through actions such as clicking and scrolling," the initial usage treats it as a known category. A brief parenthetical explanation (e.g., "an agent that interacts with a browser interface") at first use is necessary.

Second, the paper frequently uses **"scrollytelling"** (Section 5, "Analytical genres amplify..."). This is a niche term from the data visualization community. Replacing it with "scroll-driven storytelling" or "interactive scrolling narratives" would improve accessibility without losing meaning.

Third, the **"Inspector"** is a core contribution, yet it is introduced in the Abstract and Section 1 as a proper noun without a functional definition for the lay reader. The text assumes the reader understands it is a UI panel or a specific agent role immediately. A phrase like "a transparency panel (the 'Inspector') that..." would bridge this gap.

Fourth, the metric **"angle coverage"** (Section 4) is used repeatedly. While the mathematical definition follows, the term itself is jargon. "Narrative angle overlap" or "topic coverage similarity" would be more intuitive.

Finally, the term **"cross-family verifier"** (Section 4) appears in the context of using a different model family to verify claims. This is a specific technical constraint that should be briefly explained (e.g., “a verifier from a different model family to prevent bias”) rather than left as a black-box term.

These changes are minor but essential for ensuring the paper’s contributions are accessible to the broader data journalism and general AI communities, not just those deeply embedded in the current agentic workflow literature.

Action items.

- [**writing**] The manuscript relies heavily on specialized terminology and coined phrases that may alienate non-specialist readers, particularly those outside the immediate field of agentic AI systems. First, the term “computer-use agent” is introduced in the Abstract and Section 1 without definition. While the authors later describe it as an agent that “perceives the rendered interface through actions such as clicking and scrolling,” the initial usage treats it as a known category. A brief parenthetical expl

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent framework for the Data Journalist Agent, but there are specific logical inconsistencies in the quantitative reporting and metric definitions that undermine the internal consistency of the results section.

First, in Section 5.1 (“Distribution of article composition”), the authors state that the agent uses “1.45x as many sentences” but each sentence is “0.77x” the length of human sentences. Simple arithmetic (1.45 * 0.77   1.116) suggests the agent’s total word count should be approximately 11.6% *higher* than the human baseline. However, the text immediately reports the agent’s total writing volume as 1305 words versus 1557 for humans, which is a ~16% *decrease*. These three data points (sentence count ratio, sentence length ratio, and total word count) are mathematically inconsistent. The reader cannot simultaneously accept all three as stated.

Second, in Section 5.3 (“Verifiability analysis”), the definition of the metric shifts between the agent and human baselines. For the agent, “verifiability” is defined as the existence of a “traceable binding” (93% of claims). For humans, the text states the verifier has to “guess” a reproduction, and 25% of claims “recover such a binding.” This conflates the *existence* of a provenance trail with the *success* of a heuristic guess. If a human article has a claim that is factually true but lacks a code link, the agent’s metric would score it as 0 (no binding), while the human baseline metric seems to score it as 1 if the verifier successfully guesses the underlying data. This creates an apples-to-oranges comparison: the agent is measured on *traceability* (a structural property), while humans are measured on *recoverability* (a probabilistic property of the verifier). The conclusion that the gap reflects “availability of machine-checkable provenance” is logically sound only if the human metric is also strictly structural, which the text does not support.

Finally, in Section 5.2, the claim that the agent judge "preserves the human ranking" while scoring both groups "higher in absolute terms" requires a non-linear transformation assumption to hold. If the agent judge adds a constant bias to all scores, the ranking is preserved. If it adds a multiplicative bias or a bias that varies by article complexity, the ranking could change. The text asserts the preservation of ranking (rho=0.44) as a direct consequence of the higher scores, but the correlation coefficient (0.44) is actually quite low, suggesting significant rank disagreement. The text's phrasing implies a stronger agreement than the statistic supports, creating a tension between the qualitative claim ("preserves the ranking") and the quantitative evidence (rho=0.44).

Action items.

- [**science**] In Section 5.1, the stated ratios (1.45x sentences, 0.77x length) mathematically imply ~12% more total words, contradicting the reported 16% fewer words (1305 vs 1557).
- [**science**] In Section 5.3, the verifiability metric conflates 'traceability' (agent) with 'guessing success' (human). The 25% human score measures successful reconstruction, not the existence of a link, making the 93% vs 25% comparison logically invalid.
- [**writing**] In Section 5.2, claiming the agent judge 'preserves ranking' while citing a low correlation (rho=0.44) is contradictory. A rho of 0.44 indicates significant rank disagreement, not preservation.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper significantly overreaches in its claims regarding the agent's capabilities and the interpretation of its evaluation results.

First, the claim of "discovery" in Section 3.1 is unsupported. The authors state the agent "autonomously discovers an original angle" on 2026 datasets (e.g., arXiv submissions, FIFA 2026). However, the paper provides no ground-truth verification that these findings are factually correct or genuinely novel. The agent generates a narrative and visualizations, but without external validation (e.g., expert review of the specific 2026 statistics), asserting "discovery" is an extrapolation beyond the data. The system demonstrates *generation*, not necessarily *discovery*.

Second, the interpretation of the "Insight Value" scores in the human study (Section 5.1, Figure 5a) is flawed. The rubric explicitly defines a score of 3 as "Precise common knowledge" and caps scores for updates meaningful only to lay readers at 5. The agent's higher average scores likely indicate it produces generic, safe summaries that align with common knowledge, rather than providing the "non-trivial cognitive update" required for high insight. The paper conflates "consistency with common knowledge" with "high insight value," overclaiming the agent's ability to generate novel insights.

Third, the central claim that the "Inspector" makes articles "auditable" (Abstract, Section 5.3) overstates the system's capability. The evaluation measures "provenance coverage" (93% of claims have a traceable link), not "factual correctness." A claim can be perfectly traceable to a line of code that contains a bug, or to a hallucinated URL. Equating "traceability" with "auditability" or "trustworthiness" is a logical leap. The paper admits the verifier checks the *existence* of a link, not the *validity* of the underlying data or code, yet the conclusion implies a level of factual assurance that the data does not support.

Finally, the assertion that the agent "reasons about what its readers will want to read" (Abstract, Section 3.1) is an unsupported anthropomorphism. The system employs a Designer role that selects tools based on prompts; there is no evidence of actual user modeling, preference learning, or cognitive simulation. The paper extrapolates from "generating multimodal assets" to "reasoning about reader desires" without data supporting the latter.

Action items.

- [**science**] The claim that the agent 'discovers original findings' on 2026 datasets (Sec 3.1) overreaches the evidence. The paper provides no ground-truth verification that these 'discoveries' (e.g., the specific arXiv submission counts or FIFA heat-risk correlations) are factually correct or novel, only that the agent generated a narrative around them. Without external validation, 'discovery' is an unsupported extrapolation of 'generation'.
- [**science**] The conclusion that the agent 'outperforms' humans on 'Insight Value' (Sec 5.1, Fig 5a) is an over-claim. The rubric explicitly caps scores for 'common knowledge' (score 3) and 'lay intuition' (score 5). The agent's higher scores likely reflect a tendency to generate generic, safe summaries that align with common knowledge rather than providing the 'non-trivial cognitive update' required for high scores. The paper conflates 'consistency with common knowledge' with 'high insight value'.
- [**science**] The assertion that the 'Inspector' makes the article 'auditable' (Abstract, Sec 5.3) overstates the capability. The paper admits the verifier checks 'provenance coverage' (93% vs 25%), not 'factual correctness'. A claim can be perfectly traceable to a line of code that contains a bug or a hallucinated source URL. Equating 'traceability' with 'auditability' or 'trustworthiness' is a logical leap not supported by the data.
- [**writing**] The claim that the agent 'reasons about what its readers will want to read' (Abstract, Sec 3.1) is an unsupported anthropomorphism. The system uses a Designer role with tool calls; there is no evidence of actual user modeling or preference learning. The paper extrapolates from 'generating multimodal assets' to 'reasoning about reader desires' without data supporting the latter.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a novel framework for automated data journalism but requires specific clarifications regarding ethical compliance and safety considerations before acceptance.

First, the human evaluation component (Section 4.2, "Rubric evaluation & Human as judge") involves 53 participants recruited via Prolific. The manuscript currently lacks a statement regarding Institutional Review Board (IRB) approval or ethical exemption status. Given that the study involves human subjects providing subjective ratings on articles covering potentially sensitive topics (e.g., abortion access in Table 1, row 11; political settlements in Table 1, row 4), explicit confirmation of ethical oversight is mandatory for publication. The authors should add a sentence in Section 4.2 or the Appendix confirming IRB approval or exemption.

Second, while the paper emphasizes "verifiability," it does not sufficiently address the safety risks associated with generating narratives on sensitive societal issues. The evaluation set includes data on abortion clinics, political conflicts, and public health crises. There is no discussion of how the system prevents the generation of harmful, biased, or misleading content in these domains. For instance, if the agent hallucinates a statistic regarding abortion access or misrepresents political settlement data, the "verifiable" nature of the code does not guarantee the factual correctness of the underlying data or the neutrality of the narrative. A dedicated paragraph in the Discussion (Section 6) or Limitations addressing these safety risks and potential mitigation strategies (e.g., human-in-the-loop for sensitive topics) is necessary.

Finally, the use of "computer-use agents as judges" (Section 4.2) introduces a secondary layer of automation. The paper does not detail any safety protocols to ensure these agents do not generate or propagate harmful content while navigating and evaluating the articles. While this is a methodological detail, ensuring the safety of the evaluation pipeline itself is part of responsible AI research.

These issues are primarily related to research ethics and safety governance rather than the core technical novelty. Addressing them will ensure the work meets the ethical standards required for publication in a venue focused on AI and society.

Action items.

- [**science**] The human study (n=53) lacks explicit IRB/ethics approval documentation. Given the use of Prolific for human subjects and the collection of subjective ratings on sensitive topics (e.g., abortion clinics, political settlements), the manuscript must state the IRB approval number or confirm exemption status to comply with ethical research standards.
- [**writing**] The evaluation dataset includes articles on sensitive societal issues (e.g., abortion clinic access, political settlements, COVID-19 under-reporting). The paper does not address potential harms if the agent generates misleading or biased narratives on these topics. A discussion on safety guardrails or limitations regarding sensitive domains is required.
- [**writing**] The 'Computer-use agent as judge' section describes using agents to navigate and score articles. The paper does not clarify if these agents were tested for generating harmful content or if there are safeguards against the agent producing toxic or biased outputs during the evaluation process.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the central claims of the Data Journalist Agent is currently insufficient to warrant acceptance, primarily due to methodological flaws in the evaluation design and a lack of statistical rigor in the reported results.

First, the human evaluation study (n=53) reports p-values (e.g., $p<.001$ in Figure 5b) but fails to specify the statistical tests employed or report effect sizes. In a paired comparison design, the choice between a parametric (paired t-test) and non-parametric (Wilcoxon signed-rank) test depends on the distribution of the score differences, which is not discussed. Furthermore, without effect sizes (e.g., Cohen's d), the reported mean differences (e.g., +1.49 on Transparency) cannot be interpreted in terms of practical significance. A large p-value with a small effect size might indicate a statistically significant but trivial difference, while a moderate p-value with a large effect size could be meaningful. The current presentation obscures the strength of the evidence.

Second, the "Verifiability" metric (Section 5.4) introduces a severe selection bias that invalidates the comparison between agent and human articles. The agent achieves a 93% pass rate because the "Inspector" explicitly provides the code-line bindings required for the verifier to check the claim. In contrast, the human baseline achieves only 25% because the verifier is forced to "guess" the reproduction logic from the text alone, as human articles do not ship code. This metric measures the *availability of a machine-readable audit trail* rather than the *factual correctness* of the claims. It is possible that human articles are factually accurate but lack the specific metadata the verifier requires, while agent articles might be factually incorrect but possess the correct metadata structure. The paper conflates "auditability" with "verifiability," leading to a misleading conclusion about the agent's superiority in truthfulness.

Third, the use of a "computer-use agent as judge" (Section 5.3) as a proxy for human judgment is not sufficiently validated. While a correlation of rho=0.44 is reported, this is a moderate correlation that leaves significant variance unexplained. Moreover, the agent systematically assigns higher absolute scores to both groups than human judges, suggesting a calibration issue or a different interpretation of the rubric. The paper does not demonstrate that the agent's internal reasoning aligns with human judgment beyond this aggregate correlation, nor does it address the risk of the agent hallucinating its own evaluation of the interactive elements.

Finally, the claim that the agent can "discover" original findings on new data (Section 3.3) is supported only by three qualitative case studies. There is no systematic evaluation of the agent's ability to generate novel, correct insights on a held-out dataset. Without a quantitative measure of novelty or a ground-truth comparison of the insights generated, this claim remains an anecdotal demonstration rather than a scientifically supported result.

To address these issues, the authors must: (1) report the specific statistical tests and effect sizes for all human study comparisons; (2) redesign the verifiability metric to assess factual accuracy independently of the presence of an audit trail, perhaps by having human experts verify a random sample of claims from both groups; (3) provide a more robust validation of the agent-judge, including an analysis of its error modes; and (4) provide a quantitative evaluation of the “discovery” capability on a larger, held-out dataset.

Action items.

- [**science**] The human study (n=53) lacks statistical rigor. While p-values are reported, the manuscript fails to specify the statistical tests used or report effect sizes (e.g., Cohen’s d). Without these, the magnitude and reliability of the reported advantages cannot be assessed.
- [**science**] The ‘Verifiability’ metric (93% vs 25%) is methodologically biased. The agent’s score relies on explicit code-line bindings provided by the Inspector, while the human baseline is evaluated by a verifier ‘guessing’ from text. This conflates ‘auditability’ with ‘verifiability’ and unfairly penalizes human articles for lacking machine-readable trails rather than factual errors.
- [**science**] The ‘Computer-use agent as judge’ results lack validation of reliability. The correlation (rho=0.44) is moderate, yet the agent scores both groups higher than humans. The paper does not prove the agent’s rubric aligns with human intent or address potential hallucination in the agent’s scoring process.
- [**science**] The claim that the agent ‘discovers’ original findings (Sec 3.3) is anecdotal. Three examples are presented without systematic evaluation of novelty or correctness against ground truth. There is no quantitative measure of how often the agent identifies non-trivial insights versus generic summaries.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in Section 5 requires clarification regarding the choice of tests and the handling of the experimental design.

First, the manuscript reports p-values (e.g., $p < .001$) for differences in rubric scores (Section 5.1, Figure 5.1) and claim coverage (Figure 5.1b) but does not specify the statistical tests employed. Given the ordinal nature of the 1–7 Likert scale used in the rubric, non-parametric tests (e.g., Wilcoxon signed-rank test for paired data) are often more appropriate than parametric t-tests, though t-tests are sometimes used as approximations. The authors must explicitly state which test was used for each comparison. Furthermore, with multiple comparisons across five rubric dimensions and three source categories, a correction for multiple testing (e.g., Bonferroni or Benjamini-Hochberg) is necessary to control the family-wise error rate or false discovery rate. The absence of this information makes the reported significance levels difficult to validate.

Second, the experimental design described in Section 4.2 assigns each of the 53 reviewers to a single paired article (one human, one agent). This creates a paired structure where the scores for a specific article are correlated. The analysis in Section 5.1 appears to treat the 53 scores as independent samples when comparing the agent mean (4.21) to the human mean (3.38). A more rigorous approach would be to treat the *article* as the unit of analysis (n=18 pairs) or to use a mixed-effects model that accounts for the random effect of the article. Analyzing the 53 reviewer scores as independent observations ignores the clustering of data within articles, potentially inflating the degrees of freedom and leading to spurious significance (Type I error).

Finally, the error bars in Figure 5.1 are labeled as "SEM" (Standard Error of the Mean). The authors must clarify the denominator for this calculation. Is the SEM calculated across the 53 individual reviewer scores (treating them as independent), or is it calculated across the 18 article-level means? Given the study design, the latter is the statistically correct unit of inference. If the former was used, the error bars are misleadingly narrow and the associated p-values are likely invalid.

Please revise Section 5 to explicitly state the statistical tests used, the correction method for multiple comparisons, and the unit of analysis for the error bars and significance testing.

Action items.

- [**science**] Section 5.1 reports p-values (e.g., $p<.001$) for differences in rubric scores and coverage metrics but omits the specific statistical tests used (e.g., paired t-test, Wilcoxon signed-rank) and the correction method for multiple comparisons across the 5 dimensions and 3 sources. Please specify the test statistic, degrees of freedom, and whether FDR or Bonferroni correction was applied.
- [**science**] The human study (n=53) assigns each reviewer to only one paired article (Section 4.2). The analysis treats the 53 scores as independent samples for the agent vs. human comparison. However, since the data is paired by article (each article has one human score and one agent score), a paired statistical test is required to account for the variance between articles. The current analysis may inflate significance by ignoring the article-level clustering.
- [**science**] Figure 5.1 and Section 5.1 report means with 'SEM' error bars. For the human study, the unit of analysis is the reviewer, but the scores are aggregated per article. Clarify if the SEM is calculated across the 53 reviewers (treating them as independent) or across the 18 articles (treating articles as the unit of analysis). The latter is statistically more rigorous given the study design.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling narrative with a generally clear and engaging style, effectively communicating the novelty of the Data Journalist Agent. However, there are several instances of minor grammatical errors and awkward phrasing that detract from the overall polish and readability.

Specifically, in the Introduction (sec/1_intro.tex), the sentence “nearly all statistic, visual asset, and factual claim is grounded” contains a subject-verb agreement error; “statistic” should be pluralized to “statistics” to match the list and the verb “are” (or the singular “is” if rephrased). Similarly, in the Experiments section (sec/5_experiments.tex), the phrase “1 (2%) calling it a tie” is grammatically incomplete; it should be revised to “1 (2%) called it a tie” or “1 (2%) rated it a tie” for clarity.

In the Method section (sec/3_method.tex), the use of a slash in “auditability / traceability” is slightly informal for this context; replacing it with “and” or “or” would improve the flow. Additionally, in the Evaluation section (sec/4_evaluation.tex), the phrase “We provide full details information” is redundant and should be simplified to “We provide full details” or “We provide detailed information.”

While these issues are minor and do not obscure the scientific meaning, addressing them would significantly enhance the professional quality of the writing. The rest of the manuscript flows well, with clear transitions and a logical structure.

Action items.

- [**writing**] In sec/1_intro.tex, the phrase ‘nearly all statistic’ is grammatically incorrect; it should be ‘nearly all statistics’ or ‘nearly every statistic’ to agree with the singular/plural context.
- [**writing**] In sec/5_experiments.tex, the sentence ‘1 (2%) calling it a tie’ contains a grammatical error; it should be ‘1 (2%) calling it a tie’ is awkward, better phrased as ‘1 (2%) called it a tie’ or ‘1 (2%) rated it a tie’.
- [**writing**] In sec/3_method.tex, the phrase ‘auditability / traceability’ uses a slash where a conjunction like ‘and’ or ‘or’ would be more appropriate for formal academic prose.
- [**writing**] In sec/4_evaluation.tex, the phrase ‘We provide full details information’ is redundant; it should be ‘We provide full details’ or ‘We provide detailed information’.